

NLP ANSWER SHEET

Q1. What is POS Tagging? Why is it important?

POS (Part-of-Speech) tagging is a core task in Natural Language Processing (NLP). It involves assigning grammatical categories or tags to each word in a sentence. These tags represent the syntactic role of words such as noun, verb, adjective, adverb, pronoun, etc.

POS tagging helps machines understand sentence structure and meaning.

Key Concepts in POS Tagging:

1. Parts of Speech:

These are categories like nouns, verbs, adjectives, adverbs, and pronouns that define the role of a word in a sentence.

2. Tagging:

The process of assigning a specific part-of-speech label to each word in a sentence.

3. Corpus:

A large collection of text data used to train POS taggers.

Types of POS Tagging:

1. Rule-Based POS Tagging:

Uses handcrafted linguistic rules to assign tags.

2. Stochastic POS Tagging:

Uses probabilistic models such as Hidden Markov Models (HMM) to predict tags.

3. Transformation-Based POS Tagging:

Combines rule-based and statistical approaches.

4. Neural POS Tagging:

Uses deep learning and neural networks for tagging.

Why is POS Tagging Important?

1. Syntactic Analysis:

POS tags help in understanding the grammatical structure of a sentence, which is essential for parsing.

Example:

Sentence: "The cat sat on the mat."

POS Tags:

The/DT cat/NN sat/VBD on/IN the/DT mat/NN

2. Semantic Understanding:

Knowing the role of a word in a sentence helps in understanding meaning.

Example:

Sentence: "I saw the man with the telescope."

POS tags help clarify ambiguity in sentence meaning.

3. Disambiguation:

Many words in natural language are ambiguous and can belong to multiple parts of speech depending on context.

Q2. What is Named Entity? Explain NER.

Named Entity Recognition (NER) is a technique in Natural Language Processing (NLP) used to identify and classify named entities in text into predefined categories.

Examples of named entities include:

- Person names**
- Organizations**
- Locations**
- Dates**
- Time**
- Currency**
- Percentages**

NER helps machines extract structured information from unstructured text.

Definition:

Named Entity Recognition (NER) is the process of locating and classifying entities mentioned in text into categories such as person, organization, location, date, and monetary value.

Types of Named Entities:

1. Person (PER):

Names of individuals.

Example:

Elon Musk, Virat Kohli

2. Location (LOC):

Names of places.

Example:

India, Mumbai, Paris

3. Organization (ORG):

Names of companies or institutions.

Example:

Google, Microsoft, ISRO

4. Date and Time:

Specific dates and time expressions.

Example:

21st March, Monday

5. Numerical Entities:

Money, percentages, quantities.

Example:

50 dollars, 80 percent

Working of NER:

1. Text Analysis:

The system analyzes the input text.

2. Sentence Segmentation:

The text is divided into sentences.

3. Tokenization and POS Tagging:

The text is broken into tokens and grammatical tags are assigned.

4. Entity Detection and Classification:

Tokens are classified into categories like person, organization, or location.

5. Model Training:

Machine learning models improve recognition accuracy.

Applications of NER:

- Information extraction**
- Search engines**
- Chatbots**
- Question answering systems**
- Text summarization**

Q3. Applications of NLP.

Natural Language Processing (NLP) is a branch of Artificial

Intelligence that enables computers to understand, interpret, and generate human language.

Applications of NLP:

1. Chatbots:

Chatbots simulate conversation with users through text or voice.

Example:

Customer support bots.

Working:

- User sends a query.**
- NLP analyzes sentence structure and meaning.**
- System identifies user intent.**
- Chatbot generates an appropriate response.**

2. Autocomplete in Search Engines:

Autocomplete predicts search queries while users type.

How NLP Works:

- Analyzes frequently searched queries
- Understands language patterns
- Predicts relevant suggestions

Example:

Typing "weather in" may suggest "weather in Mumbai".

3. Voice Assistants:

Voice assistants interact using spoken language.

Examples:

Google Assistant, Amazon Alexa, Apple Siri.

4. Language Translation:

Automatically converts text from one language to another.

Example:

English to Hindi translation.

Popular Example:

Google Translate.

5. Sentiment Analysis:

Determines emotional tone of text.

Classification:

- Positive**
- Negative**
- Neutral**

Example:

"This phone is amazing."

6. Grammar Checking:

Grammar checkers detect and correct grammatical errors.

Examples:

- Grammarly**
- Microsoft Word**

Incorrect:

"She go to school yesterday."

Correct:

"She went to school yesterday."